BARANINATHAN S P

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SUMMARY

Computer Science Engineering student with hands-on experience in Artificial Intelligence, Machine Learning, Deep Learning, and Computer Vision. Experienced in developing real-time AI solutions using Python, OpenCV, TensorFlow, YOLOv8, and Flask. Built projects involving real-time flood detection, medical image classification, and weather forecasting. Strong understanding of machine learning workflows, data preprocessing, model development, evaluation, and deployment.

EDUCATION

FRANCIS XAVIER ENGINEERING COLLEGE — 2023 – 2027
B.E - COMPUTER SCIENCE ENGINEERING — CGPA : 7.8

LITTLE FLOWER MATRICULATION HIGHER SECONDARY SCHOOL — 2022 – 2023
MATHS COMPUTER SCIENCE GROUP — HSC - Class XII

SKILLS

Languages & Web Technologies: C, Python, SQL (MySQL), HTML5, CSS3 , JavaScript, React.js, REST APIs, Flask.
Computer Vision: OpenCV, YOLOv8, Image Processing, Object Detection, Image Classification, Video Analytics
Machine Learning: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Data Preprocessing, Model Training.
Tools: Visual Studio Code, MySQL Workbench, GitHub, Grafana, Microsoft Excel, Google Colab.
Database: MySQL, XAMPP, SQL

Portfolio | Github | LinkedIn | Leetcode

PROJECTS

AI-URBAN FLOOD DETECTION SYSTEM – AI / WEB — JAN - FEB 2026

Patented

- Team Leader & Developer: Built a patented YOLOv8 computer vision system for real-time flood monitoring.
- Developed a real-time computer vision system using Python, YOLOv8, OpenCV, and Flask for flood detection using CCTV and monitoring.

DISEASE DETECTION SYSTEM - AIML / WEB — FEB - APR 2026

- Team Leader & Developer: Developed a machine learning model utilizing Python, OpenCV, and TensorFlow.
- Developed a machine learning/deep learning model for automated medical image classification using Python, OpenCV, and TensorFlow.

AI-POWERED SMART CCTV SURVEILLANCE SYSTEM - AIML / WEB — SEP - OCT 2026

- Developed real-time vehicle entry/exit, fire detection, human counting, and industrial safety monitoring using YOLOv8 and OpenCV.
- Integrated live CCTV feeds with a Flask dashboard for security monitoring and instant alerts.

EXPERIENCE

EDGEMATRIX MULTINATIONAL CORPORATION, MADURAI — JAN - MAR 2026

- AI/ML Intern: Developed and optimized Artificial Intelligence and machine learning solutions using Python.
- Built interactive dashboards to monitor real-time data and extract actionable insights.
- Collaborated with cross-functional teams to prepare reports and improve overall project outcomes.

SOFTWARE DEVELOPER & AI CCTV SOLUTIONS (FREELANCE) — PRESENT

- Developed real-time monitoring dashboards with live camera feeds, activity tracking, security alerts, and client-focused digital solutions.
- Built AI-powered CCTV solutions for vehicle entry/exit tracking, fire detection, human detection and counting, and industrial safety monitoring.

AWARDS

- 1st Prize (15,000 cash) VISAI 2026 (16th International Project Competition)
- Published a Patent for AI-Urban Flood Detection System
- 1st Place – AATRAL Expo for Best Product
- 3rd Prize – Astro Aerospace Pvt. Ltd. Competition

CERTIFICATIONS

Red Hat Linux (RH124) | Certified in OpenCV Bootcamp | SQL by Hacker Rank | Cyber Security and Privacy in NPTEL